Growth Case: What our players are worth, and how to spend next quarter

Take-home case · Growth and User Acquisition

Your role	You have just joined as the growth lead for our mobile games.
The main question	What is a new player worth to us, and how should we spend Game A's advertising budget over the next 12 weeks?
What you hand in	A 2-page memo, an appendix, two filled-in templates, your code or spreadsheet, and your AI chat transcripts. Full list
Time	Plan for **6–8 hours** of work, spread over 5 days. Choosing what to focus on is part of the case.
Tools	Anything you like, including AI assistants. Please send us the chat transcripts.

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1. A note from your manager

From: Head of Growth · **To:** You · **Subject:** Welcome, and your first job

Hi,

Welcome to the team. Today is **Tuesday 30 September 2025**. Next quarter starts on **Monday 6 October**, and I need your recommendation for how we spend our advertising budget for Game A over the 12 weeks to **28 December**.

Our leaders don't agree on what we should do:

- **Our CFO** says: "Every quarter's advertising must earn back what it cost within 120 days. If several plans do that, pick the one that makes the most money."
- **Our CEO** says: "Growth matters. Spend as much as we can while still meeting the CFO's rule. And many of the players who find us on their own only come *because* our ads are running, so count them too."

Before I take a recommendation to them, I need you to answer three questions:

1. **What is a new player worth to us, in Game A and in Game B?** Give it by network and by country, at day 120 and after a full year. Our data has errors in it, so find them and handle

them before you trust any number. I also want to know whether a model built on one game can be trusted on the other.

2. **How should we spend Game A's budget next quarter?** One plan that meets the CFO's rule and, within that, gives the CEO as much growth as possible. Only in the networks and countries we already buy in.
3. **What ads should we build for each network next quarter, and how many?** Base it on what the data shows, and on how fast our ads wear out, so it can carry the spend plan.

One firm rule: we keep **all four of Game A's ad networks running**. The other planning rules are in Appendix A.

Everything we have is attached. Please keep the memo short. The CFO reads the first five lines, and sometimes only those.

Thanks,
Head of Growth

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2. About the games

Game A is a casual merge puzzle game, available on Android and iPhone around the world. It makes money in two ways:

- **Ads inside the game.** Players watch video ads (sometimes in exchange for a reward), and see full-screen ads and small banners.
- **In-app purchases (IAP).** Some players pay real money for items or progress.

Game B is a casual tile-matching puzzle game. It earns almost all its money from ads.

We find new players for both games in two ways:

- **Paid advertising.** We pay ad networks to show our ads in other apps. A person sees an ad, installs the game and starts playing.
  - Game A buys on four networks: **Net-A, Net-B, Net-C and Net-D**.
  - Game B has used those four and two more: **Net-E and Net-F**.

  The networks are described in Appendix B.
- **Organic.** People find the game by themselves, for example by searching the app store.

New to this world? Every term is explained where it first matters, and in the glossary. You do not need prior advertising experience.

3. What you have

All the files below are in the same folder as this document. Click a name to open it.

- All data is **as of today, 30 September 2025**.
- Installs cover **6 January to 28 September 2025** (266 days). The last two days aren't in yet.
- Money is in US dollars.
- The files are too large to work through by hand, so use Python, R, SQL or a capable spreadsheet tool.

Start with the **Data Dictionary** (PDF). It explains every column. Read it carefully: some definitions matter.

Game A

Exhibit	File	What it contains	Rows
A1	GameA_ua_daily.csv	Daily ad spend and installs, by platform, country, network and campaign.	32,773
A2	GameA_cohorts.csv	What each day's new players did afterwards: how many kept playing (up to day 120), and how much money they brought in (up to day 180).	29,165
A3	GameA_telemetry_daily.csv	The game's own tracking: daily active players, ad views and revenue, purchases, playing time, by country.	21,863
A4	GameA_telemetry_retention.csv	How many new players came back on later days, according to the game's own tracking.	674
A5	GameA_creatives_daily.csv	Spend, ad views, clicks, installs and early player value for each ad (creative).	44,387
A6	GameA_creatives_catalog.csv	List of ads: type (video, playable and so on) and the dates each was active.	32

Game B

Same file layout as Game A.

Exhibit	File	What it contains	Rows
B1	GameB_ua_daily.csv	Same as A1.	9,484
B2	GameB_cohorts.csv	Same as A2.	10,054
B3	GameB_telemetry_daily.csv	Same as A3.	15,980
B4	GameB_telemetry_retention.csv	Same as A4.	1,043
B5	GameB_creatives_daily.csv	Same as A5.	21,185
B6	GameB_creatives_catalog.csv	Same as A6.	235

Templates to fill in

For	File	What to do
Q1	Template_predictions.csv	Eight predictions for Game A, each with a range. Instructions are in the Data Dictionary.
Q2	Template_plan.csv	Your spend plan for Game A: how the quarter's budget is split across networks and countries.

4. The questions

Each question below has the same layout:

- **The question**: exactly what we are asking you to answer.
- **Key idea**: the concept you need, with a small made-up example. Examples explain ideas; they are not clues about the data, and they don't tell you which method to use. Choosing and building methods is up to you.
- **What to do**: the steps we expect to see.
- **Hand in**: what to include, and where.

QUESTION 1 · ABOUT 3–4 HOURS

What is a new player worth, in Game A and in Game B?

THE QUESTION

For each game, and for every network and country (or group of countries) in its data, how much money does a newly acquired player bring in compared with what they cost? Answer at two points in time:

- by **day 120** after install;
- after **one year** (365 days).

Do those players pay back their cost within 120 days, only within a year, or not at all?

The data contains errors. Find them, flag them, and make sure your predictions handle them properly. Don't just delete what looks odd; decide what each problem means for the numbers.

Then test whether your approach carries from one game to the other:

1. Build a model on Game A's data and predict Game A.
2. Apply that same Game A model to Game B.
3. Build a new model on Game B's data and predict Game B.
4. Apply the Game B model to Game A.

Compare the results. Where do they differ, by how much, and why? What does this tell us about using one game's model for another game?

Every spending decision rests on these numbers. If the data behind them is wrong, or the model doesn't fit the game, the plan built on it will be wrong too.

KEY IDEA 1: THE SAME PLAYERS, COUNTED BY DIFFERENT SOURCES

Three kinds of source describe the same players:

- **The ad network**: what it charged us, and the installs it believes it caused.
- **Our measurement partner (MMP)**: a neutral tool inside the game that decides which network, if any, gets credit for each install, and tracks what those players earn. This is our standard for installs and player value.
- **The game's own tracking ("telemetry")**: sees everything players do in the game, but not which ad brought them in.

These sources count in different ways, so their numbers will not match exactly. Before you trust a number, know which source it came from and how that source defines it.

> **Example:** a network reports 1,200 installs yesterday; the MMP credits it with 1,000. Both can be right: the network counts installs that another network also claims. The MMP gives each install to only one network.

A **cohort** is the group of players who installed on the same day from the same source. A cohort's "day-30" figure can only exist once the cohort is at least 30 days old.

KEY IDEA 2: PAYING BACK THE COST

- **CPI** (cost per install) = what we spent ÷ number of installs.
- **ROAS at day N** (return on ad spend) = the money a cohort has brought in by day N ÷ what that cohort cost. A cohort **pays back** when its ROAS reaches 100%.

> **Example:** Cohort X cost $4.00 per install and brings in $4.40 per player by day 120, so it pays back (110%). Cohort Y cost only $0.50 per install but brings in $0.40, so it never pays back (80%). **A cheap install is not automatically a good install, and an expensive one is not automatically bad.** What matters is how much a player brings in compared with what they cost.

KEY IDEA 3: ALWAYS LOOK COUNTRY BY COUNTRY

Countries differ a lot: installs can cost ten times more in one country than in another. When a network starts or stops buying in a country, its overall numbers move even if nothing else changed.

Example: a network buys only in the US at $4.00 per install. Then it adds a cheap country at $0.50. The network's overall cost per install drops by two-thirds, yet no country got cheaper. Only the mix changed.

That's why we work by **cell**: one network, on one platform, in one market. A market is a country or a group of countries.

KEY IDEA 4: PREDICTING VALUE, NEAR AND FAR

LTV (lifetime value) up to day N is the total money an average player has brought in by day N. Money keeps arriving for months after install, more slowly as time goes on:

Days since install	1	7	30	60	120	180	365
Money per player so far (example)	$0.25	$0.80	$1.50	$1.90	$2.30	$2.45	$2.70

For recent players you can only see the start of this curve, so day-120 value has to be predicted. The 1-year value is harder: no cohort in our data is followed for more than 180 days, so part of every year must be estimated. Long-range estimates can be far off, which makes it important to be honest about how sure you are.

A **model** here is simply your method for turning what you can see today into those predictions. We don't prescribe one. We will judge yours on four things:

- it only uses information you would really have today;
- you have checked how accurate it is on players whose outcome is already known;
- it gives honest ranges;
- someone else could re-run it and get the same answer.

WHAT TO DO

1. **Check the data first, for both games.** Find the errors and anything misleading. For each problem:
   - say what it is and show the evidence in numbers;
   - say which figures it would distort;
   - say how you handled it in your predictions: corrected, excluded, or kept on purpose.
2. **Choose your cells** for each game (network × platform × country or country group) and explain how you grouped countries.

3. For each cell, show:
    - recent weekly spend (the average over 1–28 September, which we call the **baseline**; for networks we have stopped, say when they stopped);
    - cost per install;
    - ROAS at day 7 and day 30, from cohorts you consider reliable.
4. **Build your Game A model** and predict, for each Game A cell and as ranges:
    - ROAS at day 120;
    - 1-year LTV and 1-year ROAS.

    Also give totals by network and by country group. Show how accurate the model is where the outcome is known, and explain how you estimated the part of the year the data can't show.
5. **Apply the Game A model unchanged to Game B** and record the predictions.
6. **Build a Game B model** from Game B's data, predict Game B, and then apply it to Game A.
7. **Compare all four sets**:
    - A on A;
    - A on B;
    - B on B;
    - B on A.

    Where are the differences biggest? What causes them? Which parts of a model can be shared between games, and which must be rebuilt for each game?
8. For each game, sort the cells into three groups:
    - those that **pay back within 120 days**;
    - those that pay back **only within a year**;
    - those that **don't pay back even in a year**.
9. Fill in Template_predictions.csv (Game A only): eight predictions, each with a range you are 80% confident in. We will score them against the actual outcomes, both for accuracy and for whether your ranges were honest.

Hand in, in the appendix:

- a **data problems** table: problem · where · evidence · what it distorts · how you handled it;
- a **value table for each game**: cells, plus totals by network and by country, with day-120 and 1-year figures;
- a **model comparison** table (the four sets) and half a page on what you learned;
- a short description of your models and how accurate they were;
- the filled-in prediction template.

QUESTION 2 · ABOUT 2 HOURS · MOST IMPORTANT

How should we spend Game A's budget next quarter?

THE QUESTION

Give us **one spend plan** for Game A, from 6 October to 28 December. It must:

- meet the CFO's rule;
- within that rule, give the CEO as much growth as possible.

Plan only for the **networks and countries we already buy in**; don't add new countries. Show how you would split the quarter's budget across networks and countries. Tell us:

- what it should deliver;
- how you balanced profit (CFO) against growth (CEO);
- what could go wrong.

Say whether you counted organic players towards the CFO's rule, and why.

This is the decision the CEO and CFO are waiting for. Use your values from Question 1 and the rules in Appendix A.

KEY IDEA: THE NEXT DOLLAR EARNS LESS THAN THE AVERAGE DOLLAR

The first dollars you spend reach the people most likely to install and play. Spend more, and the network has to reach people who are harder to convince and often play less. So each install tends to **cost more**, and each new player tends to be **worth less**. That's why the **average** return can hide what the **next** dollar brings back.

> **Example:** a cell earns $11,000 by day 120 from $10,000 of spend (110%). Double the spend to $20,000 and it earns $16,700. The average is still a respectable 83%, but the *extra* $10,000 only brought in $5,700 (57%).

The team's working assumptions about how strongly each network behaves like this are in Appendix A.

WHAT TO DO

1. For each network, platform and country, set the **total spend for the quarter** and its share of the budget, following every rule in Appendix A. You may group small countries, but name the

countries in each group. Also give totals by network and by country.

2. Give the **result you expect** for the quarter's players, each as a range:
   - installs;
   - money and ROAS by day 120;
   - money and ROAS after one year.
3. Explain **how you balanced the CFO and the CEO**: how much growth you gave up for profit, or the other way round, and which rules limit the plan.
4. State whether you **counted organic players**, and what in the data supports your choice.
5. Where a cell misses the 120-day rule but pays back within a year, say what you did with it and why.
6. Show at least one place where you gave **more money to a cell with a higher cost per install**, or explain why you didn't.
7. Assume each cell gets enough fresh creatives to hold its baseline CPI. Question 3 works out what that takes.
8. Name the **biggest risk** to the plan.

Hand in:

- **In the memo**: the plan and its expected result.
- **In the appendix**: the plan table (by network and country, with totals) and your reasoning.
- **The plan** in Template_plan.csv.

QUESTION 3 · ABOUT 1–1.5 HOURS

What ads should we build for each network next quarter?

THE QUESTION

Using Game A's ad-level data and the wear-out rule in Appendix A, tell us:

- **what kinds of ads to build for each network**, and why;
- **how many new creatives each network needs** next quarter to carry your spend plan from Question 2, and **when** they need to be ready;
- a **creative plan for the quarter** that puts this together.

KEY IDEA 1: WHAT GETS INSTALLS IS NOT ALWAYS WHAT GETS VALUE

An ad is called a **creative**. It can be a video, a **playable** (a small interactive demo of the game), an image, or a combination. Networks show more often the ads that turn views into installs cheaply. The usual measure is **IPM**, installs per 1,000 ad views. A high IPM makes installs cheaper, but it says nothing about *how valuable* those players are.

> **Example:** Ad P gets 30 installs per 1,000 views, so each install costs $2.00, but players bring in only $0.50 in their first week (25% back). Ad Q gets 12 installs per 1,000 views, so each install costs $5.00, but players bring in $2.25 in their first week (45% back). The network may prefer P. Q is the better business.

Networks also differ in what kinds of ads they accept and how much detail they report (Appendix B).

KEY IDEA 2: CREATIVES WEAR OUT, SO VOLUME FOLLOWS SPEND

The more often people see the same ad, the less they respond, so its CPI creeps up. More spend means more impressions, and more impressions means creatives wear out faster. That's why a bigger budget needs more new creatives, delivered on time.

> **Example:** a creative starts at $2.00 per install and gets 10% more expensive for every million impressions. It is retired at $2.60 (+30%), so it lasts 3 million impressions. A cell that will serve 12 million impressions in the quarter needs about four creatives' worth of life. If the plan doubles that cell's spend, it needs about eight.

WHAT TO DO

1. For each network, show what the data says: which types of ad get the most spend, and which bring in the most valuable players.
2. From your Question 2 plan, work out how many **impressions** each network will serve next quarter. Then use the wear-out rule to work out **how many new creatives** it needs and **by when**, remembering that today's creatives only carry the first two weeks.
3. Recommend **what to build for each network**: the type of ad, how many, and the evidence behind each choice.
4. Lay out the **creative plan for the quarter**: what gets built, for which network, and ready by which week.
5. Say what happens to CPI and to your plan's results if creatives arrive late or you build too few.

Hand in: one page in the appendix, including a table: network · impressions next quarter · new creatives needed · ready by · type of ad.

5. What to hand in

□ **memo.pdf**, **at most 2 pages**, written for the CEO and CFO. Start with these five lines:

□ your spend recommendation;

□ the result you expect, as a range;

□ which cells pay back by day 120, which only within a year, and which not at all;

□ the data problem that most affected your numbers;

□ the biggest risk.

□ **appendix.pdf**, at most 12 pages: the supporting work for Questions 1–3.

□ **predictions.csv**: your filled-in Template_predictions.csv.

□ **plan.csv**: your filled-in Template_plan.csv.

□ **Your code, notebook or spreadsheet.** We will re-run it, so it must work from the original files.

□ **AI log, half a page.**

□ Two moments when an AI tool gave you something wrong or misleading: what it said, what you did instead, and how you noticed.

□ One moment when it really saved you time.

Attach the full chat transcripts (exports or share links).

□ **A short list of what you skipped**, and why.

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6. How we will look at your work

- **Do you find the bad data, and handle it properly?** You spot the errors, flag them, and your predictions deal with them correctly.
- **Do you break things down before concluding?** Especially by country.
- **Are your predictions honest, near and far?** A clear method, checked against known results, with ranges that are realistic for day 120 and for a year.
- **Do you understand what transfers between games?** You explain why the models differ, not just that they do.
- **Is your plan sound?** It follows the rules and makes the profit-versus-growth trade-off clear.
- **Is it clear?** A busy CFO understands your memo in two minutes.
- **Do you use AI with judgment?** You catch its mistakes instead of passing them on.

7. Ground rules

- **If something is unclear**, make a sensible assumption and write it down. Deciding what to assume is part of the case, so please don't send us questions about the data.
- **AI tools are welcome.** We are not measuring how much you used them. We are looking at your judgment about when to trust them.
- **Confidential.** Please don't share these materials, and delete them when the process ends.
- **To submit**, send everything as one zip file to the address in your invitation.

Appendix A · Planning rules and assumptions (Game A)

Rule	Details
Plan period	12 weeks, from Monday 6 October to Sunday 28 December 2025.
Total budget	At most **$250,000** over the 12 weeks.
Where you can spend	Only the networks, platforms and countries Game A already buys in. No new countries. Inside Net-A you may move money between countries, and you may stop a country, but explain why.
Starting point ("baseline")	For each cell, its average weekly spend over the four full weeks 1–28 September 2025. Calculate it from GameA_ua_daily.csv.
Keep every network running	Each of these five lines must spend **at least 40% of its baseline** over the quarter, meaning at least 40% of 12 × its baseline weekly spend: Net-A Android, Net-B Android, Net-B iPhone, Net-C Android and Net-D Android.
Timing	Assume each cell's quarter spend is spread evenly across the 12 weeks.
Holiday season	In the four weeks starting Monday 24 November, installs cost **25% more** everywhere. Players are worth the same as usual. Reflect this in your expected results.
The CFO's rule	Taken together, the players we pay for in the quarter must bring in at least what they cost (ROAS ≥ 100%) by day 120 after install. • Count revenue as it is recorded in the cohort data, with in-app purchases at the full price the player paid. • You decide which ad-revenue figure is the right one; tell us which you used. • Whether organic players count is disputed (see the email).

Spending more or less (the team's rough estimates)

Spending more in a cell makes each extra dollar work less hard; spending less makes it work harder. The table shows what happens to a cell's ROAS when its average weekly spend in the quarter (quarter spend ÷ 12) is doubled or halved compared with its baseline.

Network	Double the spend → ROAS	Halve the spend → ROAS
Net-A (any country)	−21%	+27%
Net-B (Android and iPhone)	−24%	+32%
Net-C	−27%	+37%
Net-D	−36%	+57%

- In between, the change is smooth.
- Below half, there is no further gain.
- Above double is not allowed.
- For installs, assume about 70% of the ROAS change comes from installs getting more (or less) expensive.

If you want a formula: ROAS at spend multiple x = baseline ROAS × x^{-k}, with k = 0.35 (Net-A), 0.40 (Net-B), 0.45 (Net-C) and 0.65 (Net-D).

Creatives wear out

An ad gets less effective the more it is shown. For planning, use this rule:

- A **new creative** starts at the cell's baseline CPI.
- For **every 1 million impressions** a creative has served, its CPI rises by the percentage in the table (in a straight line).
- When its CPI is **30% above where it started**, it is retired and stops serving.
- A network spreads a cell's impressions evenly across the creatives that are live.
- Impressions follow spend: impressions = spend ÷ CPM × 1,000, using each cell's baseline CPM. You can calculate CPM from GameA_ua_daily.csv.
- Assume today's creatives are nearly worn out: they carry the **first two weeks** of the quarter, and after that every impression must come from new creatives.
- If a network runs out of fresh creatives, its CPI keeps rising at the table's rate. As long as you replace creatives in time, assume the cell keeps its baseline CPI.
- There is **no limit** on how many creatives the team can build. Your job is to work out how many are needed, for which network, and when.

Network (what counts as one creative)	CPI rise per 1 million impressions	So a creative is retired after about
Net-A (each video or playable we upload)	+20%	1.5 million impressions
Net-B (each ad set; one set serves both Android and iPhone)	+300%	100,000 impressions
Net-C (each ad pack)	+300%	100,000 impressions
Net-D (the listing image)	+150%	200,000 impressions

Why so different? In-app networks charge far more per impression, so the same budget buys far fewer impressions, and each creative is seen by a smaller, more engaged audience that tires of it sooner.

Appendix B · The ad networks

Network	What it is	Ads it takes	How much detail it reports
Net-A	A large self-service ad platform. We set the budget, target and countries; its system decides where the ads appear (video sites, search results, other apps).	We upload pieces (videos, images, playables, text) and it builds the ads itself.	By type of placement (display, search, video), not by individual ad.
Net-B	An ad network that shows our ads inside other mobile games and apps.	Ad "sets": usually a few videos plus a playable and/or an end screen.	By ad set.
Net-C	A video ad network inside other mobile games.	Video followed by a playable, or a playable alone.	By ad pack.
Net-D	A rewards app. Its users browse a list of games and earn rewards for playing, and spending in, the games they install.	Only a listing image in their app.	Very little at ad level.
Net-E, Net-F	Game B only. Net-E is an in-app ad network; Net-F is a social media ad platform.	Individual videos, images and playables.	By individual file or ad.
Organic, Store-Search, Unattributed	• **Organic**: no ad got credit for the install. • **Store-Search**: credited to a search inside the app store. • **Unattributed**: the measurement partner couldn't tell.		

Appendix C · Glossary

Term	Meaning
Install	A new player opening the game for the first time.
Cohort	All players who installed on the same day from the same source.
Cell	One network, on one platform, in one market (a country or group of countries). Our unit of analysis and planning.
CPI	Cost per install = spend ÷ installs.
Retention (day N)	The share of a cohort that plays again on day N after installing.
LTV	Lifetime value: total money per player brought in up to a given day. "1-year LTV" is the total after 365 days.
ROAS (day N)	Return on ad spend: money a cohort brought in by day N ÷ what it cost. 100% means it has paid back.
Pay back	The point where a cohort has brought in as much as it cost.
Model	Your method for turning what you can see today into predictions of future value.
IPM	Installs per 1,000 ad views.
Impression	One showing of an ad.
Creative wear-out (fatigue)	An ad getting less effective, so its CPI rises, the more it has been shown.
CPM	Cost per 1,000 ad views. CPI = CPM ÷ IPM.
CTR	Share of ad views that were clicked.
IAP	In-app purchase: real money a player spends inside the game.
MMP	Mobile measurement partner: the neutral tool that decides which ad gets credit for each install.
Mediation SDK	Software inside our game that picks which ad network fills each ad slot, and reports the ad money we earn.

Organic	A player who found the game without an ad getting credit.
Paid vs blended ROAS	Paid ROAS counts only players we paid for. Blended ROAS adds the money from organic players against the same ad spend.